GURKEERAT JAWANDA

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EDUCATION

University of Canberra *Canberra, ACT*
Graduated
December 2025

Bachelorsof SoftwareEngineering Major:
Robotics and ArtificialIntelligence
GPA: - 6.1/7DeansExcellenceAwardee

ACADEMIC & RESEARCH EXPERIENCE

Capstone Research Project

Power Grid Blackout Prediction

- Designed and developed an end-to-end machine learning pipeline to predict power grid blackout risks across 17 generators using operational and system-level performance data. The project involved building a structured workflow covering data preprocessing, feature engineering, model development, validation, and performance evaluation to support reliable predictive analysis in a high-impact energy systems context.
- Applied rigorous data cleaning and transformation techniques to prepare raw generator and grid data for modelling, including handling missing values, scaling features, and constructing meaningful predictor variables to improve model robustness and stability. Trained and compared multiple classification algorithms, including Logistic Regression, Random Forest, and K-Nearest Neighbours (KNN), with systematic cross-validation and hyperparameter optimisation to identify the best-performing approach.
- Evaluated model effectiveness using a range of engineering-focused performance metrics such as Accuracy, Precision, Recall, F1-Score, and Log Loss, ensuring a balanced understanding of predictive performance beyond simple accuracy. Interpreted results in the context of real-world grid reliability and failure prediction, and formally documented the methodology, findings, and technical recommendations under academic supervision. This project demonstrated strong capability in applied machine learning, critical analysis, and the development of data-driven solutions for complex infrastructure systems.

Computer Vision

Bird Species Recognition

- Developed and evaluated advanced computer vision pipelines for fine-grained bird species classification, combining both traditional feature-engineering methods and modern deep learning approaches. The project focused on recognising visually similar bird species from image datasets, requiring careful attention to feature representation, model design, and evaluation methodology.
- Implemented a hybrid classical vision pipeline based on Scale-Invariant Feature Transform (SIFT), Bag-of-Visual-Words (BoVW), and Support Vector Machines (SVM) to extract robust handcrafted image descriptors and perform species classification. In parallel, designed and tested Convolutional Neural Network (CNN)-based models to learn hierarchical visual features directly from image data, enabling a comparative study between conventional computer vision methods and deep learning architectures.
- Conducted image preprocessing, feature extraction, model training, and five-fold cross-validation to ensure the reliability and repeatability of results. Performance was assessed using class-weighted accuracy, recognition rate, and confusion matrices, allowing for a detailed understanding of classification behaviour across multiple species categories. The project clearly demonstrated improved predictive performance and feature representation when transitioning from handcrafted feature pipelines to deep learning-based solutions, highlighting practical understanding of image classification, experimental design, and model evaluation in computer vision applications.

Machine Learning For Financial Market Analysis

Stock Price Movement Prediction[Python, SQL, R]

- Engineered a modular machine learning framework for predicting daily stock price direction using financial time-series data, including Open, High, Low, Close, and Volume (OHLCV) variables. The project was designed to explore the use of machine learning in financial market analysis by building scalable and reusable workflows for classification-based forecasting.
- Developed a structured pipeline in Python, with supporting use of SQL and R for data handling, analysis, and experimentation. Performed extensive feature engineering by constructing technical and statistical indicators from raw market data to capture price momentum, volatility, and short-term market behaviour. Applied Principal Component Analysis (PCA) for dimensionality reduction, improving model efficiency while preserving key patterns in the dataset.
- Trained and evaluated multiple classification models, including Support Vector Machines (SVM), Logistic Regression, and Extreme Gradient Boosting (XGBoost), to assess their suitability for financial prediction tasks. Emphasis was placed on writing modular, reusable scripts for preprocessing, model training, testing, and future retraining, ensuring the project followed sound software engineering and machine learning engineering principles. This work demonstrated the ability to combine quantitative analysis, predictive modelling, and robust code design to address real-world problems in financial analytics.

PERSONAL PROJECTS

AI Recipe Recommender *[Python, Flask, Spoonacular API, HTML, CSS, JavaScript]*

Designed and developed a Flask-based web application that recommends recipes based on user-provided ingredients by integrating with the Spoonacular API. Built server-side functionality to process user input, manage API requests, and dynamically render recipe results using Jinja templating. Implemented a responsive and user-friendly front end using Bootstrap 5, ensuring consistent performance across devices. Structured the project using a modular file architecture to support maintainability, scalability, and efficient local deployment.

Robotics & Mechatronics *[SolidWorks, Mechanical Design, Motion Simulation]*

Created detailed 3D CAD models of engineering components and assemblies, including a Knuckle Joint, Pressure Plate, and a 4-DOF Robotic Arm, using SolidWorks. Conducted motion simulation and workspace analysis to assess mechanical feasibility and functional performance. Strengthened practical skills in precision modelling, assembly design, mechanical analysis, and technical CAD documentation.

Android App *[Java, Kotlin, Android Studio, Google ML Kit, Firebase Realtime Database]*

Developed an Android application featuring real-time image analysis capabilities, including text recognition, image labelling, and barcode scanning using Google ML Kit Vision APIs. Integrated Firebase Realtime Database to store recognised text, image metadata, and user-edited outputs for persistent cloud-based data management. Applied a modular MVVM architecture with RecyclerView, ViewModel, and LiveData to support scalable development, clean separation of concerns, and efficient user interface updates.

CERTIFICATIONS

- DeepLearning.AI - Generative AI Specialisation
- OpenCV University - Computer Vision and YOLO
- AWS Cloud Practitioner (Planned)
- UC Davis - Data Visualisation with Tableau
- Google - Advanced Machine Learning on Google Cloud Specialization

SKILLS

Languages: Python,R,JavaScript,Java,HTML, CSS, SQL, MATLAB, Kotlin
DevOps and Tools: Git,Docker,TensorFlow, PyTorch, VS Code, OpenCV, Jupyter Notebook, Anaconda, SolidWorks
Project Management and Collaboration: Slack, Microsoft Teams, Notion, Scrum, Agile, GitHub, Markdown
CurricularActivities: TeamManager,DebateClubs,ActiveReaderinArtificialIntelligenceandMachineLearning Literature

TECHNICAL SKILLS

MLOps & Development: Model training pipelines, Experiment tracking Docker, Git, GitHub, Flask REST APIs, Model versioning and retraining concepts

Cloud & Platform: AWS(EC2, S3 - Foundational) , Azure AI Studio , Firebase Realtime Database

Robotics & Simulation: ROS, SolidWorks, Motion Simulation, Workspace Analysis

Machine Learning/AI: Machine Learning, Deep Learning, Computer Vision, CNNs, Vision Transformers (ViT), YOLO (Object Detection), Large Language Models (LLMs), Small Language Models (SLMs), Vision-Language Models (VLMs), Prompt Engineering, Supervised Fine-Tuning (SFT), LoRA, Multimodal AI (Text and Image Pipelines), Feature Engineering, PCA Model Evaluation